# Evaluation runs: 2pc-tr (two-pc-trillium)

generated 2026-11-08 from brain-data 614b86250 2026-11-08 11:41:25 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 12 | conforms | 0.0 | 4 | 1 h, 250 GB | 2 | 4 | brain03, 2026-11-02 |
| 3 | 8 | 87 | conforms | 0.0 | 4 | 1 h, 250 GB | 2 | 9 | brain03, 2026-11-02 |
| 4 | 16 | 888 | conforms | 0.1 | 4 | 1 h, 250 GB | 2 | 17 | brain03, 2026-11-02 |
| 5 | 32 | 16365 | conforms | 0.4 | 4 | 1 h, 250 GB | 2 | 27 | brain03, 2026-11-02 |
| 6 | 64 | 531648 | conforms | 8.4 | 4 | 1 h, 250 GB | 2 | 20 | brain03, 2026-11-02 |
| 7 | 128 | 25499439 | conforms | 222.8 | 46 | 1 h, 250 GB | 2 | 57 | brain03, 2026-11-02 |
| 8 | 256 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 4 | violation | 0.0 | 4 | 1 h, 250 GB | 6 | 19 | brain03, 2026-11-02 |
| 3 | 8 | 96 | 54 | violation | 0.0 | 4 | 1 h, 250 GB | 56 | 435 | brain03, 2026-11-02 |
| 4 | 16 | 1560 | 960 | violation | 0.8 | 4 | 1 h, 250 GB | 962 | 15371 | brain03, 2026-11-02 |
| 5 | 32 | 39120 | 24600 | violation | 13.9 | 4 | 1 h, 250 GB | 24602 | 787206 | brain03, 2026-11-02 |
| 6 | 64 | 1409040 | 889920 | violation | 428.9 | 16 | 1 h, 250 GB | 889922 | 56954928 | brain03, 2026-11-02 |
| 7 | 128 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 7 ignore-abort-vote bag, 8 bag correct, 8 correct bag, 8 bag ignore-abort-vote, 8 bag correct.t16.p.r3, 8 bag ignore-abort-vote.t16.p.r3, 9 bag ignore-abort-vote, 9 bag correct.

skipped: a smaller size had reached its limit: 8 ignore-abort-vote bag, 9 ignore-abort-vote bag, 9 correct bag, 10 correct bag, 10 ignore-abort-vote bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 4 | 12 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 8 | 87 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 16 | 888 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 32 | 16365 | conforms | 0.5 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 64 | 531648 | conforms | 9.5 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 7 | 128 | 25499439 | conforms | 866.8 | 46 | 48 h, 250 GB | brain05, 2026-11-02 |
| 8 | 256 | 1626502608 | conforms | 17768.0 | 1208 | 48 h, 250 GB | brain04, 2026-11-07 |

### stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | conforms | 0.0 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 3 | 8 | 66 | conforms | 0.0 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 4 | 16 | 728 | conforms | 0.1 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 5 | 32 | 15240 | conforms | 0.4 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 6 | 64 | 523632 | conforms | 8.2 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 7 | 128 | 25438672 | conforms | 258.0 | 51 | 48 h, 250 GB | brain03, 2026-11-07 |
| 8 | 256 | 1626000000 | conforms | 17818.0 | 1200 | 48 h, 250 GB | brain03, 2026-11-07 |

### stateful: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 4 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 8 | 96 | 54 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 16 | 1560 | 960 | violation | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 32 | 39120 | 24600 | violation | 1.6 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 64 | 1409040 | 889920 | violation | 21.7 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7 | 128 | 69048000 | 43641360 | violation | 1201.2 | 132 | 48 h, 250 GB | brain05, 2026-11-02 |
| 8 | 256 | 4419112320 | 2793369600 | violation | 67427.0 | 4864 | 48 h, 250 GB | brain05, 2026-11-02 |

### stateful: pair **ignore-abort-vote**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 4 | violation | 0.0 | 4 | 48 h, 250 GB | brain03, 2026-11-07 |
| 3 | 8 | 96 | 54 | violation | 0.0 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 4 | 16 | 1560 | 960 | violation | 0.2 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 5 | 32 | 39120 | 24600 | violation | 1.7 | 2 | 48 h, 250 GB | brain03, 2026-11-07 |
| 6 | 64 | 1409040 | 889920 | violation | 41.7 | 27 | 48 h, 250 GB | brain03, 2026-11-07 |
| 7 | 128 | 69048000 | 43641360 | violation | 1355.4 | 118 | 48 h, 250 GB | brain03, 2026-11-07 |
| 8 | 256 | 4419112320 | 2793369600 | violation | 57992.0 | 4838 | 48 h, 250 GB | brain03, 2026-11-07 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 8 bag ignore-abort-vote.t16.p.r3, 10 correct bag, 10 bag ignore-abort-vote, 10 ignore-abort-vote bag, 11 ignore-abort-vote bag, 11 correct bag, 12 ignore-abort-vote bag, 12 correct bag, 13 ignore-abort-vote bag, 13 correct bag, 14 correct bag, 14 ignore-abort-vote bag, 15 correct bag, 15 ignore-abort-vote bag.

left out by the table rules (ok): 7 ignore-abort-vote bag, 7 correct bag.

left out by the table rules (timeout): 8 ignore-abort-vote bag, 8 correct bag, 9 bag ignore-abort-vote, 9 bag correct.

skipped: a smaller size had reached its limit: 9 ignore-abort-vote bag, 9 correct bag, 10 ignore-abort-vote bag, 10 correct bag.

the tool ended with an error (exit-101): 10 bag correct.

left out by the table rules (invalid: foreign load): 11 bag correct, 11 bag ignore-abort-vote.

## offline

### offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 12 | conforms | 0.0 | 8 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 8 | 87 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 16 | 888 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 32 | 16365 | conforms | 0.8 | 8 | 10 min, 16 GB | 0.0 | 0.8 | 0.0 | in-memory | brain05, 2026-11-02 |
| 6 | 64 | 531648 | conforms | 10.9 | 28 | 1 h, 250 GB | 0.0 | 10.9 | 0.0 | merge | brain05, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 128 | 25499439 | conforms | 318.9 | 46 | 1 h, 250 GB | 0.1 | 318.9 | 0.0 | merge | brain05, 2026-11-03 |
| 8 | 256 | 1626502608 | conforms | 19373.0 | 1208 | 48 h, 250 GB | 0.0 | 19373.0 | 0.0 | merge | brain04, 2026-11-07 |

### offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | conforms | 0.3 | 6 | 48 h, 250 GB | 0.3 | 0.0 | 0.0 | merge | brain03, 2026-11-07 |
| 3 | 8 | 66 | conforms | 0.0 | 8 | 48 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain03, 2026-11-07 |
| 4 | 16 | 728 | conforms | 0.1 | 8 | 48 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain03, 2026-11-07 |
| 5 | 32 | 15240 | conforms | 0.4 | 8 | 48 h, 250 GB | 0.1 | 0.4 | 0.0 | merge | brain03, 2026-11-07 |
| 6 | 64 | 523632 | conforms | 10.6 | 28 | 48 h, 250 GB | 0.0 | 10.6 | 0.0 | merge | brain03, 2026-11-07 |
| 7 | 128 | 25438672 | conforms | 246.6 | 46 | 48 h, 250 GB | 0.1 | 246.6 | 0.0 | merge | brain03, 2026-11-07 |

### offline: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 2 | violation | 0.0 | 8 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 8 | 96 | 6 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 16 | 1560 | 14 | violation | 0.3 | 8 | 10 min, 16 GB | 0.0 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 32 | 39120 | 30 | violation | 2.1 | 10 | 1 h, 250 GB | 0.0 | 2.0 | 0.1 | merge | brain05, 2026-11-03 |
| 6 | 64 | 1409040 | 62 | violation | 29.3 | 30 | 1 h, 250 GB | 0.0 | 29.3 | 0.0 | merge | brain05, 2026-11-03 |
| 7 | 128 | 69048000 | 126 | violation | 1114.3 | 122 | 1 h, 250 GB | 0.1 | 1114.3 | 0.0 | merge | brain05, 2026-11-03 |

### offline: pair **ignore-abort-vote**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 2 | violation | 0.3 | 6 | 48 h, 250 GB | 0.3 | 0.0 | 0.0 | merge | brain03, 2026-11-07 |
| 3 | 8 | 96 | 6 | violation | 0.0 | 8 | 48 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain03, 2026-11-07 |
| 4 | 16 | 1560 | 14 | violation | 0.1 | 8 | 48 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain03, 2026-11-07 |
| 5 | 32 | 39120 | 30 | violation | 1.9 | 8 | 48 h, 250 GB | 0.1 | 1.9 | 0.0 | merge | brain03, 2026-11-07 |
| 6 | 64 | 1409040 | 62 | violation | 33.6 | 30 | 48 h, 250 GB | 0.0 | 33.6 | 0.0 | merge | brain03, 2026-11-07 |
| 7 | 128 | 69048000 | 126 | violation | 994.9 | 120 | 48 h, 250 GB | 0.1 | 994.7 | 0.2 | merge | brain03, 2026-11-07 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 fifo correct, 8 fifo ignore-abort-vote, 8 bag ignore-abort-vote.p.r3, 9 bag ignore-abort-vote, 9 bag correct.

skipped: a smaller size had reached its limit: 8 correct bag, 8 bag ignore-abort-vote, 8 ignore-abort-vote bag, 9 ignore-abort-vote bag, 9 correct bag, 10 correct bag, 10 ignore-abort-vote bag.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 12 | conforms | 2.7 | 164 | 10 min, 16 GB | 2.7 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 8 | 87 | conforms | 7.1 | 157 | 10 min, 250 GB | 6.8 | 0.1 | 0.3 | merge | brain01, 2026-11-03 |
| 4 | 16 | 888 | conforms | 3.8 | 153 | 10 min, 250 GB | 3.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 2 | violation | 2.7 | 164 | 10 min, 16 GB | 2.7 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 8 | 96 | 6 | violation | 6.8 | 157 | 10 min, 250 GB | 6.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 16 | 1560 | 14 | violation | 3.9 | 153 | 10 min, 250 GB | 3.8 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag ignore-abort-vote, 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote, 9 bag ignore-abort-vote, 9 bag correct.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 12 | conforms | 2.7 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 8 | 87 | conforms | 2.9 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 16 | 888 | conforms | 3.0 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 10 | 4 | violation | 2.7 | 160 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 8 | 96 | 54 | violation | 3.0 | 162 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 4 | 16 | 1560 | 960 | violation | 3.1 | 162 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote, 9 bag ignore-abort-vote, 9 bag correct.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.1 | 157 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 4.5 | 162 | 10 min, 250 GB | 4.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 888 | 888 | conforms | 12.9 | 2015 | 10 min, 250 GB | 12.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.1 | 176 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 4.5 | 156 | 10 min, 250 GB | 4.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 1560 | 1560 | conforms | 10.0 | 559 | 10 min, 250 GB | 9.9 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 2 | violation | 3.1 | 157 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 87 | 96 | 6 | violation | 4.5 | 162 | 10 min, 250 GB | 4.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 888 | 1560 | 14 | violation | 12.9 | 2015 | 10 min, 250 GB | 12.8 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag ignore-abort-vote, 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.1 | 157 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 4.6 | 162 | 10 min, 250 GB | 0.1 | 4.5 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 888 | 888 | conforms | 12.9 | 2015 | 10 min, 250 GB | 0.1 | 12.8 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.1 | 176 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 4.6 | 156 | 10 min, 250 GB | 0.1 | 4.5 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 1560 | 1560 | conforms | 10.0 | 559 | 10 min, 250 GB | 0.3 | 9.9 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 2 | violation | 3.1 | 176 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 96 | 6 | violation | 4.5 | 156 | 10 min, 250 GB | 0.1 | 4.5 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 888 | 1560 | 14 | violation | 10.0 | 559 | 10 min, 250 GB | 0.1 | 9.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag ignore-abort-vote, 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.1 | 168 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 4.3 | 160 | 10 min, 250 GB | 4.2 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 888 | 888 | conforms | 12.8 | 2126 | 10 min, 250 GB | 12.7 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.1 | 164 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 4.5 | 172 | 10 min, 250 GB | 4.4 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 1560 | 1560 | conforms | 13.2 | 2428 | 10 min, 250 GB | 13.0 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 2 | violation | 3.1 | 168 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 96 | 6 | violation | 4.2 | 160 | 10 min, 250 GB | 4.2 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 888 | 1560 | 14 | violation | 12.7 | 2126 | 10 min, 250 GB | 12.7 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag ignore-abort-vote, 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.1 | 168 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 4.4 | 160 | 10 min, 250 GB | 0.1 | 4.2 | 0.2 | merge | brain01, 2026-11-03 |
| 4 | 888 | 888 | conforms | 12.7 | 2126 | 10 min, 250 GB | 0.1 | 12.7 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.1 | 164 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 4.6 | 172 | 10 min, 250 GB | 0.1 | 4.4 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 1560 | 1560 | conforms | 13.1 | 2428 | 10 min, 250 GB | 0.3 | 13.0 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 2 | violation | 3.1 | 164 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 87 | 96 | 6 | violation | 4.5 | 172 | 10 min, 250 GB | 0.1 | 4.4 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 888 | 1560 | 14 | violation | 13.0 | 2428 | 10 min, 250 GB | 0.1 | 13.0 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag ignore-abort-vote, 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.2 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 8.2 | 162 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 888 | 888 | conforms | 13.3 | 2044 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.2 | 158 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 8.2 | 170 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 1560 | 1560 | conforms | 16.7 | 2129 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 4 | violation | 3.1 | 160 | 1 h, 250 GB | 1881 | brain01, 2026-11-02 |
| 3 | 87 | 96 | 54 | violation | 8.0 | 165 | 1 h, 250 GB | 4054 | brain01, 2026-11-03 |
| 4 | 888 | 1560 | 960 | violation | 17.1 | 2116 | 1 h, 250 GB | 4494 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 12 | conforms | 3.2 | 164 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 87 | 87 | conforms | 3.5 | 168 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 888 | 888 | conforms | 7.2 | 429 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ignore-abort-vote**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 10 | 10 | conforms | 3.1 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 96 | 96 | conforms | 3.6 | 168 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 1560 | 1560 | conforms | 9.8 | 950 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ignore-abort-vote**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 10 | 4 | violation | 3.2 | 164 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 87 | 96 | 54 | violation | 3.5 | 168 | 1 h, 250 GB | 6 | brain01, 2026-11-02 |
| 4 | 888 | 1560 | 960 | violation | 7.3 | 429 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 bag correct, 8 bag ignore-abort-vote.